

# AYUSH GAGGAR

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## EDUCATION

### Northwestern University

*Ph.D. Candidate in Mechanical Engineering (Robotics)*

**2022 - 2027**

*Chicago, IL*

### University of California, San Diego

*B.S. Mechanical Engineering, Minor in Entrepreneurship & Innovation, GPA: 3.96/4.00*

**2020 - 2022**

*La Jolla, CA*

## PROFESSIONAL EXPERIENCE

### Graduate Researcher, Northwestern University

**Jan. 2023 – Present**

*Interactive & Emergent Autonomy Lab — PI: Prof. Todd Murphey*

*Evanston, IL*

- Researching time evolutions of visual features in the latent space, striving for safe control using learned world models.
- Pushing the limit for neural radiance fields (NeRFs) & Gaussian Splatting for robustness with low data.
- Previous research: linear-complexity ergodic control, dual-arm manipulation, and benchmarks on Franka 7DOF arm.
- Research interests: leveraging uncertainty as a property of robustness in machine learning, visual world models (VLMs/VLAs, latent space), and active learning in robotics and deep learning settings.

### Undergraduate Researcher, University of California, San Diego

**Jan. 2021 – Jun. 2022**

*Bioinspired Robotics Design Lab — PI: Prof. Mike Tolley*

*La Jolla, CA*

- Developed mechanics for differential body geometries for locomotion of underwater squid-inspired robot.
- Optimized cam design for power performance by varying compression to coasting ratios using Arduino and MATLAB.
- Successfully ran underwater experiments for displacement and velocity performance for varied body geometries.

### Mechanical/Thermal Engineering Intern

**Jun. 2021 – Sep. 2022**

*Advanced Micro Devices, Inc. (AMD)*

*Sunnyvale, CA*

- Designed a test fixture for EPYC server sockets to ensure pin connectivity remains within specification under static loading tests up to 50 kgf; socket models from 2019-2023 passed, providing empirical support for QA standards.
- Developed new retimer chip cooling assembly, used in > 200 servers. Developed bill of materials, cost justification, soldering, and mechanical clearance checks.
- Took initiative to prototype socket trays to expedite Genoa product launch, preventing failure of up to 50 sockets.

## PUBLICATIONS

### Journals

- [1] Max Muchen Sun, **Ayush Gagggar**, Peter Trautman, and Todd Murphey. “Fast Ergodic Search with Kernel Functions.” *IEEE Transactions on Robotics*, 2025.

### Conferences

- [1] **Ayush Gagggar**, Todd Murphey. “NeRF-Control: Perception-Aware Trajectory Optimization” *International Conference on Robotics and Automation (ICRA)*, 2026. [Under Review]
- [2] **Ayush Gagggar**, Todd Murphey. “Data Augmentation for NeRF in the Low Data Limit.” *International Conference on Robotics and Automation (ICRA)*, 2025.

### Patents

- [1] Max Muchen Sun, **Ayush Gagggar**, Peter Trautman, and Todd Murphey. “Kernel-Based Ergodic Search.” SN: 18/914929. 2024. [Under Review]

## PROJECTS

### EKF SLAM for Differential Drive Robot

**Jan. 2023 - Mar. 2023**

- Implemented Extended Kalman Filter SLAM from scratch in ROS2 C++, using LiDAR data to localize a Turtlebot3.
- Integrated odometry readings, supervised and unsupervised learning from LiDAR data, and created a simulation environment in RViz2 for the full functioning SLAM algorithm.

### Robot Play: Don't Let the Balloon Touch the Ground

**Nov. 2022 - Dec. 2022**

- Controlled a 7DOF Franka arm to track, predict, move to, and hit a balloon such that it would never hit the ground.
- Developed a ROS2 control system from scratch, including forward and inverse kinematics in cartesian and joint space.
- Employed a RealSense RGB-D camera to isolate the centroid of moving, red objects and forward predict its motion.

### 360° Scanning Sound Array

**Jan. 2022 - Jun. 2022**

- Designed a novel testing apparatus to measure noise (dB and freq.) emitted by small-scale UAVs for industry sponsor ATA Inc. as a function of drone's 3DOF moment in 3D space, primarily for national defense interests.
- Primarily manufactured using manual and CNC mills and lathes, with FEA analysis for a 4.24 factor of safety.
- Implemented open loop motor control to determine microphones' spherical positioning with an accuracy of  $\pm 0.015^\circ$ .

## TECHNICAL SKILLS

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**Software:** Python, C++, Robotics Operating System (ROS/ROS2), PyTorch, Jax, MATLAB, Linux, Git, Java  
**Deep Learning:** NeRFs, VAEs, Diffusion Models, GANs, MLPs, Bayesian Neural Networks, Gaussian Processes  
**Simulators:** MuJoCo, Gazebo, RViz, PyBullet, CoppeliaSim  
**Design:** Solidworks, Fusion 360 (CAD & CAM), Cura, Simulink, LTSpice  
**Fabrication:** FDM 3D Printing, Mill, CNC, Soldering, Instron, Optical Scanner

## AWARDS

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|-----------------------------------------------------------------------------------|------------------|
| Doctoral Consortium, <i>International Conference on Robotics &amp; Automation</i> | <b>May 2025</b>  |
| Murphy Fellowship, <i>Northwestern University</i>                                 | <b>Sep. 2022</b> |
| Magna Cum Laude, <i>University of California, San Diego</i>                       | <b>Jun. 2022</b> |
| Best Student Presenter, <i>Project in a Box, UCSD Undergraduate Conference</i>    | <b>May 2021</b>  |
| 2x Dean's and President's Scholar, <i>San José State University</i>               | <b>May 2019</b>  |
| Eagle Scout, <i>Boy Scouts of America</i>                                         | <b>Aug. 2018</b> |

## LEADERSHIP & SERVICES

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|--------------------------------------------------|----------------------------|
| <b>Private Tutor &amp; Peer Educator</b>         | <b>Nov. 2017 – Present</b> |
| <i>UCSD Dept. of Engineering &amp; Freelance</i> | <i>Multiple Locations</i>  |

- Tutored over 125 UCSD engineering students in Physics and Math, which involves hosting office hours, leading hourly, biweekly review sessions, and creating original worksheet content to help mentor students.
- Increased student engagement by using Discord and unique teaching styles by 225% as compared to other sections.
- A personal passion of mine is teaching in STEM and mentoring students.

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|----------------------------------------------------------------------|-------------------------------|
| <b>Teaching Assistant</b>                                            | <b>Fall 2024, Spring 2025</b> |
| <i>Northwestern University, Department of Mechanical Engineering</i> | <i>Evanston, IL</i>           |

- Assisted in an upper-division Dynamics class with 80 students each quarter with grading and office hours.
- Guest lectured on dynamics concepts like the Lagrangian and Special Rotation matrices.
- Overwhelmingly positive student evaluation (96%) of skill in communication and engagement.

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|-----------------------------------------|------------------------|
| <b>Museum Presenter – Volunteer</b>     | <b>Apr. 2023, 2024</b> |
| <i>Museum of Science &amp; Industry</i> | <i>Chicago, IL</i>     |

- Presented my robotics research during national robotics week, where thousands of children from grades K-12, along with adults, interacted with the experimental systems.

|                                          |                              |
|------------------------------------------|------------------------------|
| <b>Resident Advisor</b>                  | <b>Aug. 2019 - Jun. 2020</b> |
| <i>San José State University Housing</i> | <i>San Jose, CA</i>          |

- Built a community for 41 residents, which entailed publicizing events, enforcing policies, and resolving peer conflicts.